جامعة بيرزيت
BIRZEIT UNIVERSITY

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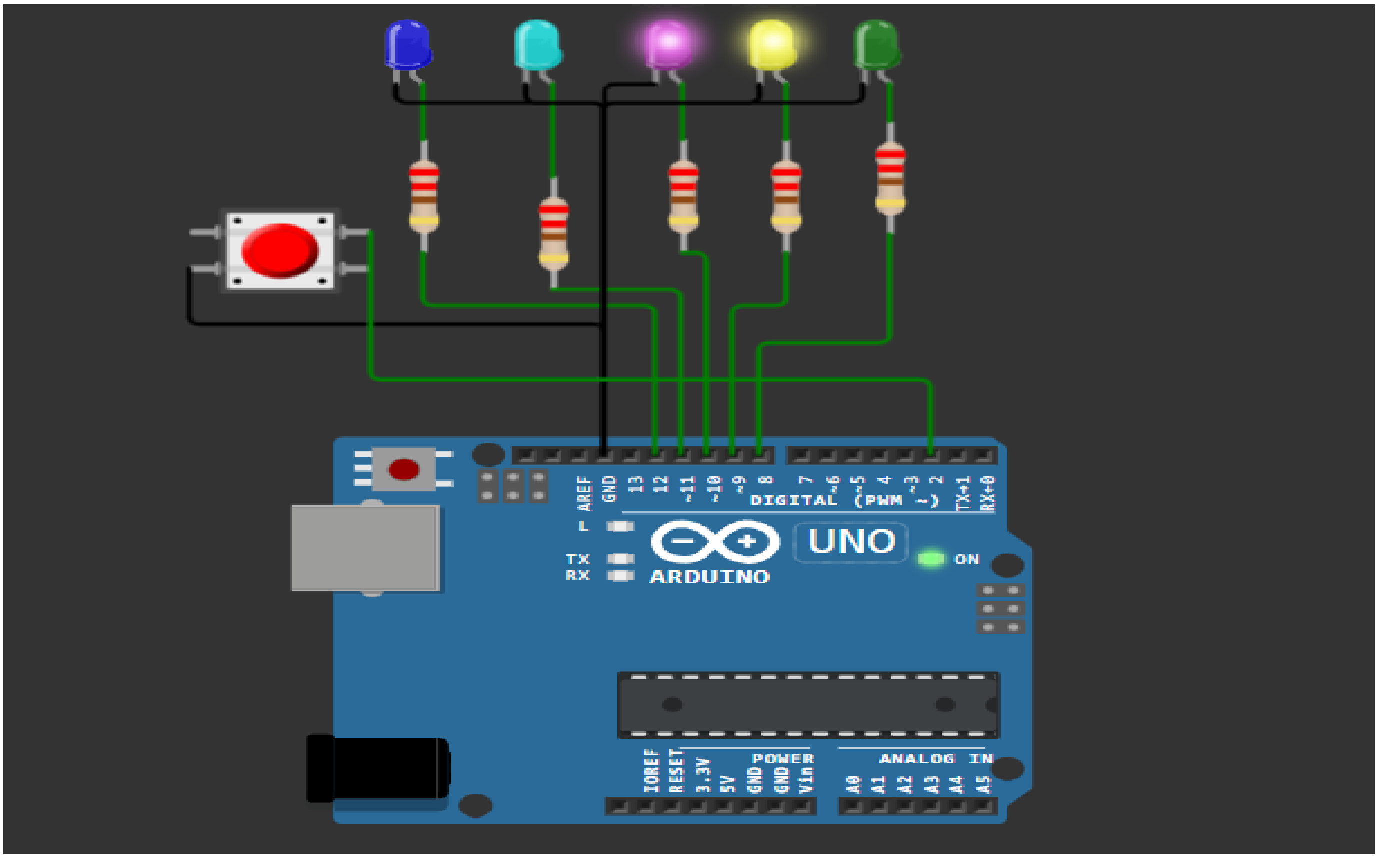
Question 2 — Non-Blocking LED Patterns with millis() (no delay)



This project uses an Arduino Uno connected to five LEDs and one push button. LEDs A, B, C, and D are connected to digital pins 8, 9, 10, and 11 through 220-ohm resistors, while the bonus LED is connected to pin 12. All LED cathodes are connected to GND. The push button is connected to pin 2 and GND, and the internal pull-up resistor is enabled in the code. The program controls the LEDs using direct register manipulation. LED A blinks every 500 ms, LED B every 300 ms, and LED C every 750 ms. LED D changes state whenever the button is pressed with debounce protection. The bonus LED turns ON only when LEDs A, B, and C are all ON at the same time. The program also sends the LED states to the Serial Monitor every 2 seconds.